Locally Equivalent Weights for Multilevel Regression and Poststratification

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Running example: Same-Sex Marriage Survey

Analysis of US state-level support for same-sex marriage[1].

- **Survey population:** Five consistently-coded national polls from 2004 ($N_S = 6,341$)
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We'll analyze this data with two methods: **calibration weighting** (CW) and **MrP** (multilevel regression and poststratification).

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Two methods

Survey: Observe $(\mathbf{x}_i, y_i)$ for $i = 1, \ldots, N_S$
Let $Y_\mathcal{S} = \{y_1, \ldots, y_{N_S}\}$.

Target: Observe $\mathbf{x}_j$ for $j = 1, \ldots, N_T$.
We want $\mu := \frac{1}{N_T} \sum_{j=1}^{N_T} y_j$, but don't observe y_j.

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Black box!

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Raw survey mean: $\frac{1}{N_S} \sum_{i=1}^{N_S} y_i = 0.333$

Raking: $\hat{\mu}^{\text{CW}}(Y_{\mathcal{S}}) = 0.345$

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How are MrP and raking are using the data differently?

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Linear and nonlinear models

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Key idea

Noting that $w_i := N_S \frac{\partial \hat{\mu}^{\text{CW}}(Y_{\mathcal{S}})}{\partial y_i}$, take $w_i^{\text{MrP}} := N_S \frac{\partial \hat{\mu}^{\text{MrP}}(Y_{\mathcal{S}})}{\partial y_i}$.

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The slope w_i^{MrP} is a *MrP locally equivalent weight*, or *MrPlew*.

It measures the effect of local perturbations to y_i on the estimator $\hat{\mu}^{\mathrm{MrP}}$.

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$$w_i^{\mathrm{MrP}} := N_S \frac{\partial \hat{\mu}^{\mathrm{MrP}}(Y_{\mathcal{S}})}{\partial y_i} = N_S \mathrm{Cov}_{\mathcal{P}(\theta|Y_{\mathcal{S}})}\left(\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\intercal \mathbf{x}_j), \frac{\partial}{\partial y_i} \log \mathcal{P}(y_i|\mathbf{x}_i, \theta) \right)$$

Can estimate w_i^{MrP} *without re-running MCMC* [Gus96; GBJ18].

Same-Sex Marriage Survey Weights

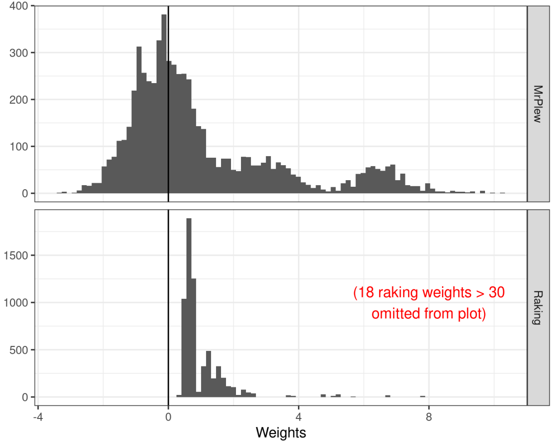

Figure 1: Comparison of weights for the Same-Sex Marriage analysis

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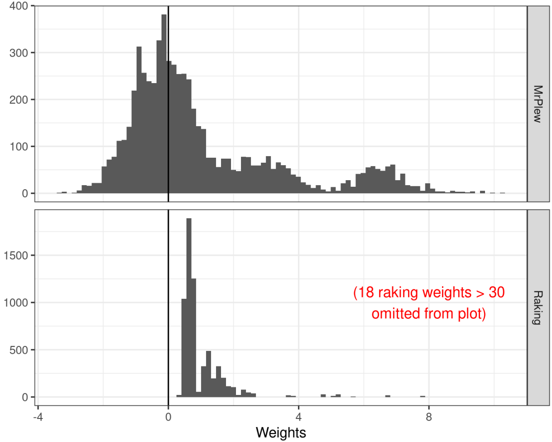

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What does the difference between these weights mean in terms of

- Frequentist sampling variance (standard errors)?
- Covariate balance / external validity?
- Subgroup contributions?

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Question:

How do responses in Texas affect our estimate of public opinion in CA?

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More specific question:

Suppose we *increased* the responses y_i in Texas.

How would our estimate of public opinion in CA change?

Take $\tilde{y}_i = y_i + \delta \mathbb{I}\left(i \text{ is in Texas}\right)$. Then $\hat{\mu}^{\mathrm{CW}}(\tilde{y}) - \hat{\mu}^{\mathrm{CW}}(y) = \delta \underbrace{\sum_{i \text{ in Texas}} w_i}_{\text{Texas's contribution}}$.

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Theorem: [Gio+26] (informal)

Take $\tilde{y}_i = y_i + \delta f(\mathbf{x}_i)$. We state conditions under which

$$\sup_{f \in \mathcal{F}} \left| \hat{\mu}^{\mathrm{MrP}}(\tilde{Y}_{\mathcal{S}}) - \hat{\mu}^{\mathrm{MrP}}(Y_{\mathcal{S}}) - \delta \sum_{i=1}^{N_S} w_i^{\mathrm{MrP}} f(\mathbf{x}_i) \right| = O(\delta^2) \text{ as } N \to \infty$$

...for a very broad class of $\mathcal{F}$. [3]

Uniformity helps justify searching over a large class of perturbations f.

[3] $\mathcal{F}$ can be any Donsker class of measurable functions with uniformly bounded $\mathbb{E}\left[\|\mathbf{x}\| \ f(\mathbf{x})\right] \ \frac{2}{2}$.

Same-Sex Marriage subgroup contribution

Summarize state contribution: $\sum_{i \text{ in state}} w_i$

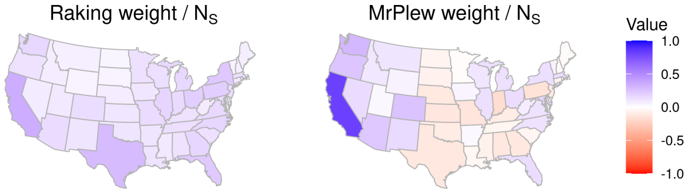

Figure 2: Subgroup contribution for Same-Sex Marriage in California

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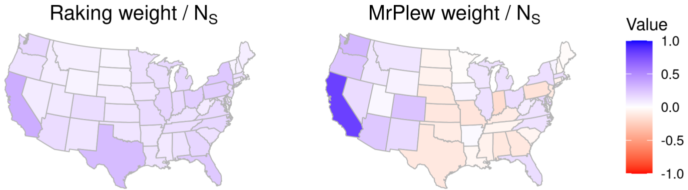

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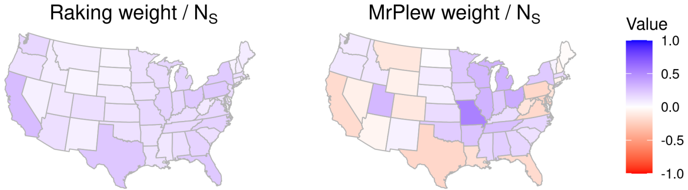

Figure 3: Subgroup contribution for Same-Sex Marriage in Missouri

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Stop! That is a misleading conclusion!

MrP is modeling the response, not the sampling bias.

CW-like checks are not diagnostics for the response model.

A correctly specified response model can do badly on CW checks, and vice-versa! (See paper.)

This work is best thought of as *model interrogation* for MrP.

Conclusion (let's try that again)

By defining *MrP locally equivlanent weights*, we can rigorously provide analogues of many CW diagnostics for non-linear MrP estimators for model interrogation.

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R package available at
`https://github.com/rgiordan/mrplew`

Preprint at
`https://arxiv.org/abs/2606.04250`

Extra slides

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Same-Sex Marriage frequentist variance

Theorem [Gio+26] (informal):

You can use the weights for frequentist variance as if MrP were linear if (and only if) y_i is a sufficient statistic of the model.

Frequentist variance of $\sqrt{N_S}\hat{\mu}^{\mathrm{MrP}}$ and $\sqrt{N_S}\hat{\mu}^{\mathrm{CW}}$
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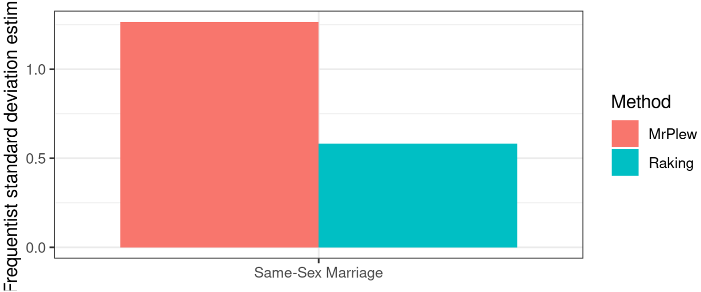

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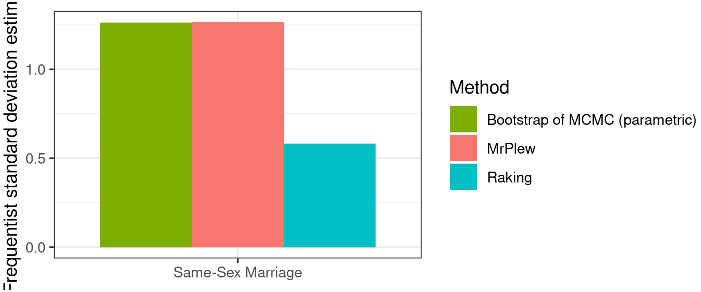

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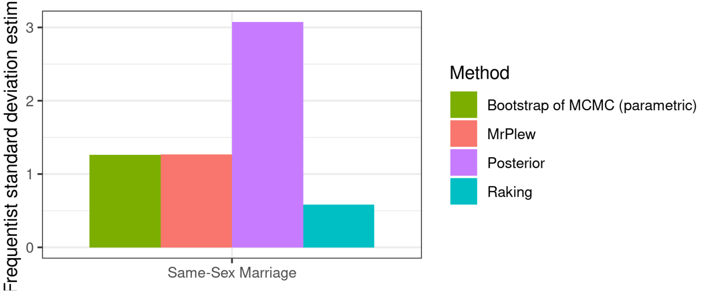

Note that frequentist and posterior variance can be very different in misspecified models with many random effects!

Same-Sex Marriage Survey Weights

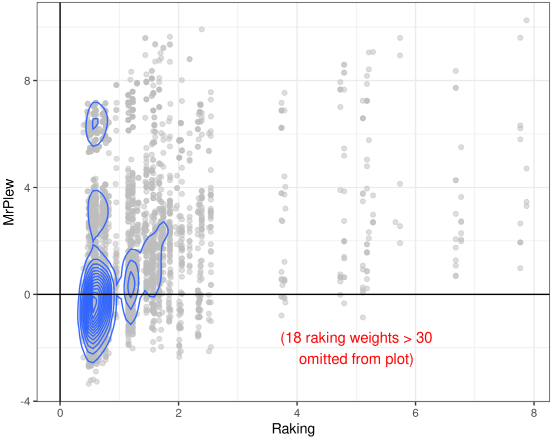

Figure 4: Comparison of weights for the Same-Sex Marriage analysis

Same-Sex Marriage covariate balance

Covariate balance / external consistency: $\frac{1}{N_T}\sum_{j=1}^{N_T} f(\mathbf{x}_j) \overset{\text{check}}{=} \frac{1}{N_S}\sum_{i=1}^{N_S} w_i f(\mathbf{x}_i)$